

When Does Conformer Geometry Help? Complementarity of 3D Ensemble Statistics and 2D Fingerprints for Molecular Property Prediction

Bryan Cheng*
bcbc7264@gmail.com
Great Neck South High School
Great Neck, NY, USA

Jasper Zhang*
jasperzhang1001@gmail.com
Great Neck South High School
Great Neck, NY, USA

Abstract

When do three-dimensional conformer ensembles improve molecular property prediction beyond two-dimensional fingerprints? We provide the first systematic, mechanistically grounded answer. Through $\sim 1,000$ experiments spanning 13 model configurations, 14 regression targets, and 2 classification targets across MoleculeNet, QM9, and MARCEL benchmarks, we discover *selective complementarity*: conformer ensemble statistics extracted via Distribution Kernel Operators (DKOs) yield statistically significant RMSE reductions on solvation-dependent properties (ESOL -11.0% , $p < 10^{-9}$; FreeSolv -13.5% , $p < 3 \times 10^{-5}$; 10-seed paired validation) while providing no benefit for electronic or steric tasks. Three lines of evidence confirm this selectivity has a physical rather than statistical basis: improvement is larger under scaffold splits than random splits ($+11.9\%$ vs. $+8.5\%$ on ESOL), concentrates on large, flexible molecules ($+18.9\%$ for heaviest quartile), and grows monotonically with training data. We establish a four-tier performance hierarchy: end-to-end 3D GNNs (SchNet, PaiNN; 21–42% over fingerprints) $\approx$ engineered physicochemical descriptors (PMI/SASA/USR) $>$ Morgan fingerprints + XGBoost $>$ all neural conformer ensemble methods, confirmed by two architecturally diverse GNNs and revealing that the pre-computed feature bottleneck limits ensemble approaches. Feature attribution and mutual information analysis expose the mechanistic asymmetry: conformer mean features carry $2\text{--}8\times$ more information per feature than fingerprint bits, yet covariance features contribute $< 2\%$ of model signal, explaining why five simple scalar invariants outperform all complex covariance architectures ($p < 0.001$). These findings yield an empirical property taxonomy and a practical decision framework for when conformer generation is worth the investment.

CCS Concepts

• **Computing methodologies** $\rightarrow$ Machine learning; Modeling and simulation; • **Applied computing** $\rightarrow$ Life and medical sciences.

*All authors contributed equally.

Keywords

molecular property prediction, conformer ensembles, distribution kernel operators, fingerprints, 3D descriptors, covariance representations

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1 Introduction

Molecular property prediction is a cornerstone of computational chemistry and drug discovery. While graph neural networks (GNNs) operating on 2D molecular graphs have achieved remarkable success [10, 35], a molecule's biological activity fundamentally depends on its three-dimensional geometry. Small molecules adopt multiple low-energy conformations in solution, and properties such as solvation free energy and lipophilicity are Boltzmann-weighted averages over this conformational ensemble [11].

Accurate prediction of solvation properties is particularly critical for ADMET (absorption, distribution, metabolism, excretion, and toxicity) profiling in early-stage drug discovery [14, 20], where experimental measurement is expensive and computational screening must efficiently prioritize candidates (Figure 1).

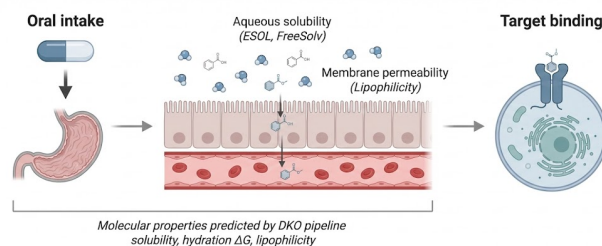


Figure 1: Mechanistic context for molecular property prediction. Oral drugs must dissolve (aqueous solubility), cross epithelial membranes (lipophilicity), and reach their cellular target. The DKO pipeline predicts these solvation-dependent properties from conformer ensemble statistics. Created with BioRender.com [3].

This has motivated a growing body of work on conformer ensemble methods [6, 32, 38] and 3D-aware GNNs [8, 30, 31], which incorporate 3D geometry for property prediction. However, a fundamental question remains unanswered: *when does explicit 3D conformer geometry provide signal beyond what 2D molecular descriptors already capture, and why?* Prior work has shown that classical descriptors often match neural methods [34] and that conformer ensembles help on steric tasks [38], but no study has systematically characterized *which property types* benefit from 3D conformer information, with mechanistic explanation and rigorous statistical validation.

We address this gap through 13 model configurations across 14 regression targets spanning three benchmark families (MoleculeNet, QM9, MARCEL). Using Distribution Kernel Operators (DKOs), we extract first-order (mean μ) and second-order (covariance Σ) statistics from conformer ensembles and evaluate multiple fusion architectures, alongside end-to-end 3D GNN baselines (SchNet, PaiNN). We investigate three concrete research questions:

- RQ1. Property taxonomy:** We characterize which molecular property types benefit from 3D conformer geometry and provide a mechanistic basis for the observed selectivity.
- RQ2. Representation hierarchy:** We establish how pre-computed conformer ensemble methods compare to 2D fingerprints and end-to-end 3D GNNs (SchNet, PaiNN) across task types.
- RQ3. Representation complexity:** We test whether complex covariance representations outperform simple summary statistics, grounded in a bias-variance analysis of parameterization complexity versus dataset size.

Scope: This study evaluates hybrid FP+conformer methods on both regression (14 targets) and classification (BACE, BBBP) tasks; end-to-end neural DKO variants are evaluated on regression only, as classification requires architectural adaptation (Section 4.8). Our key contributions are:

- (1) **Empirical property taxonomy** (Figure 4): An evidence-based framework for when conformer geometry adds value, grounded in feature attribution, mutual information analysis, and error stratification. Solvation-dependent properties benefit from hybrid features; steric/Boltzmann properties benefit from attention-based conformer weighting; electronic properties show no improvement. This provides practitioners a decision framework for when to invest in conformer generation.
- (2) **Selective complementarity with mechanistic explanation:** Hybrid FP+conformer features yield statistically significant improvements on solvation tasks (ESOL -11.0% , $p < 10^{-9}$; FreeSolv -13.5% , $p < 3 \times 10^{-5}$; 10-seed paired t -tests). Critically, ESOL improvement is *larger* under scaffold splits ($+11.9\%$) than random splits ($+8.5\%$), ruling out data leakage. The benefit concentrates on large, flexible molecules, consistent with the physical mechanism.

- (3) **Performance hierarchy:** End-to-end 3D GNNs (SchNet, PaiNN) surpass fingerprints by 21–42% on solvation tasks, while fingerprints outperform all neural conformer ensemble methods across every benchmark. Enhanced physicochemical 3D descriptors (PMI, SASA, USR) achieve comparable gains to SchNet on ESOL. Two architecturally diverse GNNs (continuous-filter and equivariant) confirm this hierarchy—not previously established—revealing that the pre-computed feature bottleneck, not the absence of 3D information, limits conformer ensemble methods.
- (4) **DKO framework:** A Distribution Kernel Operator architecture with 7 covariance representation variants, 28 enhanced 3D descriptors, and a gated fusion mechanism that significantly outperforms attention ($p < 0.001$). Five simple covariance summary statistics outperform all complex representations.

2 Related Work

The MARCEL benchmark [38] established standardized evaluation for conformer ensemble learning on Kraken steric descriptors, bond dissociation energies, and electronic properties. GeoMol [6] generates conformers via end-to-end learning, while 3D Infomax [32] uses contrastive 3D pre-training to enrich 2D representations. Alternative conformer generation methods include OMEGA [12] and CREST [24]; we use ETKDG for consistency with the MARCEL protocol. These approaches typically aggregate conformer features via mean pooling or attention, discarding distributional information.

In the 3D representation space, SchNet [30] introduced continuous-filter convolutions for 3D coordinates; DimeNet++ [7] adds directional message passing; PaiNN [31] achieves equivariant message passing; and Equiformer [19] combines equivariant attention with transformers. These methods operate on single conformers; our work instead models the *ensemble distribution* via summary statistics.

On the pre-training front, GROVER [28] uses self-supervised graph transformers on large molecular corpora. These pre-trained models [28, 37] often achieve state-of-the-art results on MoleculeNet benchmarks but require large-scale pre-training data. We do not benchmark pre-trained models, as our focus is understanding when 3D conformer geometry *per se* adds predictive value—an orthogonal question to whether pre-training on large molecular corpora helps. This scoping decision means our property taxonomy is validated in the non-pre-trained setting; pre-trained 3D models such as Uni-Mol [37] may extract useful 3D signals even for electronic properties through learned representations trained on millions of conformers, which could alter the taxonomy boundaries. We explicitly note this as a limitation (Section 6). Importantly, DKO conformer statistics are *complementary* to pre-trained representations: conformer μ/Σ features could be concatenated with pre-trained embeddings just as they are concatenated with fingerprints here, potentially compounding improvements on solvation tasks. We leave this combination for future work.

Among classical approaches, extended-connectivity fingerprints [27] remain the dominant representation in cheminformatics. Combined with gradient-boosted trees [4], they provide baselines that neural methods frequently fail to surpass [15, 34]. Jiang et al. [15] systematically showed that descriptor-based models outperform GNNs on many drug discovery tasks, a finding our work extends to the 3D conformer domain. Our work quantifies this gap and identifies where conformer features add complementary value.

Processing unordered conformer sets also requires permutation-invariant architectures. DeepSets [36] provides the theoretical foundation, while Set Transformer [18] introduces attention-based aggregation. We compare these against our DKO framework, which models the distribution’s first and second moments explicitly. More broadly, kernel methods have a long history in molecular property prediction [29]. Our DKO builds on distributional kernels that compare molecules via their conformer distributions, using a learned positive semi-definite kernel $\mathbf{K} = \mathbf{LL}^\top$.

3 Methods

Figure 2 provides an overview of the DKO pipeline and hybrid approach.

3.1 Conformer Generation and Feature Extraction

For each molecule, we generate $n = 50$ conformers using RDKit’s [16] ETKDG (Experimental Torsion-angle preference with Knowledge and Distance Geometry) algorithm [26] with energy minimization via the MMFF94 (Merck Molecular Force Field) force field. The choice of $n=50$ follows the MARCEL benchmark protocol [38]; conformer count ablations show that $n=5-10$ suffice for XGBoost-based models. Unlike MARCEL, which uses Boltzmann-weighted conformer aggregation, our pipeline computes unweighted mean and covariance statistics. This simplification reduces implementation complexity but may underweight low-energy conformers that dominate the thermodynamic ensemble, particularly for Boltzmann-averaged properties (Section 4.7).

From each conformer’s 3D coordinates, we compute a feature vector $\mathbf{x}_i \in \mathbb{R}^D$ via the following procedure: for each pair of atoms within a 4.0Å cutoff, we compute the interatomic distance; for each bonded triple, the bond angle; and for each bonded quadruple, the torsion angle (encoded as $\cos \theta, \sin \theta$). These geometric features are concatenated with 19-dimensional per-atom features (element type, hybridization, formal charge, etc.) and zero-padded or truncated to dimension D . DKO neural models use $D = 1024$. For the XGBoost hybrid, $\boldsymbol{\mu}$ is computed from features truncated to $D = 256$; a sensitivity analysis sweeping $D \in \{256, 512, 1024, 2048\}$ (Section 4.9) confirms $D = 256$ is optimal: larger D adds noise dimensions that the shallow model cannot exploit, and RMSE monotonically increases with D .

Table 1: DKO variants and baselines. $|\mathbf{s}|$: covariance summary dimension. Ablation variants marked with $\dagger$.

Model	Covariance representation	$ \mathbf{s} $
dko_gated	Learned gate fusing $\boldsymbol{\mu}/\boldsymbol{\Sigma}$	64
dko_invariants	5 scalar invariants (tr, log-det, $\ \cdot\ _F$, etc.)	5
dko_eigenspectrum	Top- k eigenvalues	64
dko_lowrank	Top- k eigenvalues + eigenvector proj.	128
dko_residual	$\boldsymbol{\mu}$ pred. + learned $\boldsymbol{\Sigma}$ correction	64
dko_crossattn	Cross-attn ($\boldsymbol{\mu}$ queries, $\boldsymbol{\Sigma}$ keys)	64
dko_router	Diversity-based MoE routing	64
dko_first_order †	$\boldsymbol{\mu}$ -only (no covariance)	0
dko_diagonal †	Per-feature variances only	1024
attention	Set Transformer [18] aggregation	—
mean_ensemble	Mean pooling over conformers	—
single_conformer	Lowest-energy conformer only	—
DeepSets	Permutation-invariant [36]	—
dko_separate_nets	Indep. $\boldsymbol{\mu}/\boldsymbol{\Sigma}$ nets (no fusion)	—

Given a conformer ensemble $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, we compute:

$$\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \in \mathbb{R}^D, \quad (1)$$

$$\boldsymbol{\Sigma} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^\top + \lambda \mathbf{I} \in \mathbb{R}^{D \times D}, \quad (2)$$

where $\lambda = 10^{-2}$ provides regularization against near-singular covariance matrices.

3.2 DKO Architecture

The Distribution Kernel Operator (DKO) maps the pair $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ to a molecular property prediction. The core architecture consists of:

Kernel construction. A low-rank factorization $\mathbf{K} = \mathbf{LL}^\top$ ($\mathbf{L} \in \mathbb{R}^{k \times D}$, $k = 64$) projects the mean into a k -dimensional subspace: $\tilde{\boldsymbol{\mu}} = \mathbf{L}\boldsymbol{\mu}$.

Covariance representation. A variant-specific function $\mathbf{s} = g(\boldsymbol{\Sigma}) \in \mathbb{R}^{|\mathbf{s}|}$ extracts a fixed-length summary, ranging from 5 scalar invariants to full eigenspectrum projections (Section 3.3).

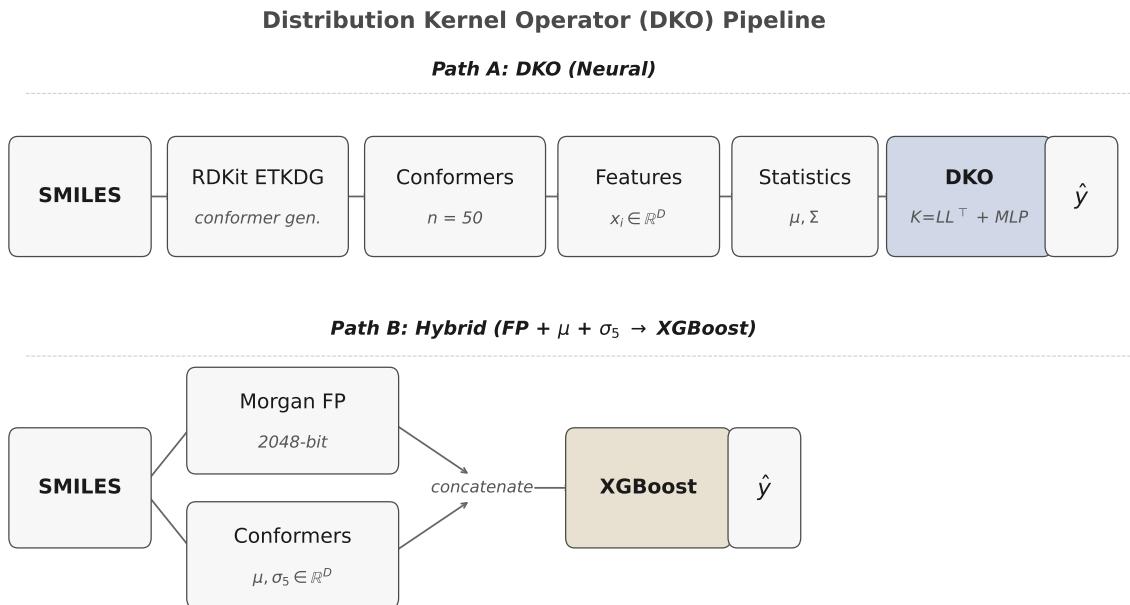
Prediction. A 3-layer MLP ($[k+|\mathbf{s}|, 256, 128, 1]$, ReLU, dropout 0.1) produces $\hat{y} = f_\theta([\tilde{\boldsymbol{\mu}}; \mathbf{s}])$, trained with MSE loss.

3.3 DKO Variants

We evaluate 9 covariance representation strategies (Table 1), all using eigendecomposition of $\boldsymbol{\Sigma}$ (with a diagonal proxy when $D > 256$).

3.4 Enhanced 3D Shape Descriptors

We implement 28 property-relevant 3D descriptors in four categories: shape (9: 2 PMI ratios, 3 raw principal moments, asphericity, eccentricity, radius of gyration, inertial shape factor), surface (1: total SASA [22]), USR (12: rotation-invariant shape signature [1]), and global geometry (6: span, molecular volume, compactness, 3 bounding box extents). These capture physically relevant 3D variation—SASA relates to solvation free energy, PMI ratios encode shape anisotropy.



μ = conformer mean, Σ = conformer covariance, σ_5 = 5 covariance summary statistics (trace, log-det, $\|\cdot\|_F$, eigenvalue ratios), D = feature dimension

Figure 2: Overview of the DKO pipeline. Left: SMILES $\rightarrow$ $n=50$ conformers (ETKDG) $\rightarrow$ geometric features ($D=1024$). Center: DKO extracts ensemble μ and Σ ; low-rank kernel $K=LL^T$ maps to predictions via variant-specific fusion. Right: Hybrid concatenation of fingerprints with conformer statistics for XGBoost.

3.5 Hybrid FP + Conformer Approach

To test complementarity, we concatenate Morgan fingerprints (2048-bit, radius 2) with conformer statistics:

$$\mathbf{z} = [\text{FP}_{2048}; \mu_d; \sigma_5] \in \mathbb{R}^{2048+d+5}, \quad (3)$$

where μ_d is the conformer mean projected to $d=256$ dimensions via PCA, and σ_5 denotes 5 variance-based summary statistics from Σ (total variance, top-5 eigenvalue variance, maximum eigenvalue, effective rank, mean eigenvalue variance). We train XGBoost [4] on these combined features.

3.6 Conformer Diversity Metric

We quantify conformer diversity per molecule as $\text{CDM} = \text{tr}(\Sigma) = \sum_{i=1}^D \lambda_i$, the total variance across all conformer features. High CDM indicates large geometric variation.

3.7 Training Details

All DKO and baseline neural models are trained with AdamW [21] (learning rate 10^{-4} , weight decay 10^{-5}), 300 epochs with early stopping (patience 30), and feature normalization along the conformer dimension only ($\text{dim} = 1$). Mixed precision is disabled for DKO variants due to numerical sensitivity of covariance arithmetic. We use 3 seeds for the main benchmark and 10 seeds for statistical validation; 10 seeds provide sufficient power to detect effect sizes of $\sim 5\%$ at $\alpha = 0.05$

given the observed variance across architectures. All datasets use Murcko scaffold-based 80/10/10 train/validation/test splits [2], where molecules sharing the same core scaffold are kept in the same split. Scaffold splits are generated by extracting the Bemis–Murcko generic scaffold [2] for each molecule using RDKit, sorting scaffolds by frequency, and assigning entire scaffold groups to train/validation/test in order until the 80/10/10 ratio is met. For the main benchmark, each model configuration is evaluated over 3 independent seeds (controlling weight initialization and data shuffling); for statistical validation experiments (Section 4.10), we use 10 seeds to provide sufficient power to detect effect sizes of $\sim 5\%$ at $\alpha = 0.05$. We additionally compare scaffold vs. random splits to quantify generalization to unseen chemical scaffolds (Section 4.11).

4 Experiments

4.1 Datasets

We evaluate on three dataset categories:

MoleculeNet regression [34]: ESOL (1,128 molecules, aqueous solubility [5]), FreeSolv (642, hydration free energy [23]), Lipophilicity (4,200, octanol–water partition coefficient), and QM9 (133,885 molecules [25]; we predict three targets: HOMO–LUMO gap, HOMO energy, and LUMO energy, denoted QM9-Gap, QM9-HOMO, QM9-LUMO).

Table 2: Summary of the 14 regression targets used in this study, grouped by property category.

Dataset	Property	N	Category
ESOL	Aqueous solubility	1,128	Solvation
FreeSolv	Hydration ΔG	642	Solvation
Lipophilicity	LogP	4,200	Solvation
QM9-Gap	HOMO–LUMO gap	133K	Electronic
QM9-HOMO	HOMO energy	133K	Electronic
QM9-LUMO	LUMO energy	133K	Electronic
BDE	Bond dissoc. energy	5,915	Electronic
Drugs-75K ($\times 3$)	Electronic props	75K	Electronic
Kraken ($\times 4$)	Sterimol B5/L/burB5/burL	1,552	Steric

Table 3: Morgan FP + XGBoost vs. best neural conformer model per dataset (mean of 3 seeds; std omitted for clarity, all $<5\%$ of mean). Neural Deficit: relative RMSE increase of neural model over FP baseline. Best neural model identified in parentheses (G = dko_gated, I = dko_invariants, A = attention, M = mean_ensemble).

Dataset	FP+XGB	Best Neural	Neural Deficit
ESOL	1.507	1.635 (G)	+8.5%
FreeSolv	2.939	4.077 (A)	+38.7%
Lipophilicity	0.910	1.131 (I)	+24.3%
QM9-Gap	0.0203	0.0364 (A)	+79.3%
QM9-HOMO	0.0142	0.0188 (A)	+32.4%
QM9-LUMO	0.0187	0.0335 (M)	+79.0%

MARCEL benchmark [38]: Kraken [9] (1,552 phosphine ligands, 4 Sterimol steric descriptors (B5, L, buried-B5, buried-L)—Boltzmann-averaged 3D properties), BDE (5,915 bond dissociation energies), Drugs-75K (75,099 drug-like molecules, 3 electronic properties).

Table 2 summarizes all targets. The benchmark spans 7 electronic, 3 solvation, and 4 steric targets. The hybrid FP+conformer improvement emerges on two of the three solvation tasks where conformational geometry directly influences the property (ESOL, FreeSolv), suggesting a genuine mechanistic link rather than a statistical artifact. Lipophilicity, while physically solvation-related, shows no hybrid improvement—consistent with its dependence on equilibrium partitioning rather than conformational flexibility.

4.2 Fingerprint Baseline

We first quantify the gap between fingerprint baselines and neural conformer methods. Table 3 compares Morgan fingerprints with XGBoost against the best neural conformer model per dataset. Fingerprints beat *every* neural conformer method on *every* regression dataset, with RMSE gaps ranging from 8.5% (ESOL) to 79.5% (QM9-Gap). This result holds across all 8 MARCEL benchmark targets as well.

4.3 Neural Model Comparison

Given the strong fingerprint baseline, we examine whether neural conformer methods can close this gap. Table 4 presents

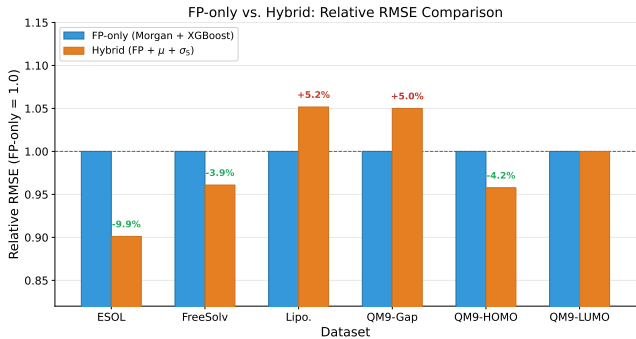


Figure 3: Relative RMSE of hybrid FP+ μ + Σ features vs. FP-only baseline (dashed line at 1.0). Values below 1.0 indicate improvement. Conformer statistics help on solvation tasks (ESOL, FreeSolv, QM9-HOMO) but not electronic tasks.

the full benchmark across 13 model configurations and 4 representative datasets. The original DKO with PCA-compressed covariance ranks 12th, motivating our eigendecomposition variants: all 7 new variants improve on average over it. Among neural methods, attention achieves the best average rank (3.25), followed by mean ensemble (4.25). The new DKO variants—particularly dko_invariants (rank 3, best on Lipophilicity) and dko_gated (rank 4, best on ESOL)—are competitive. We note that with only 3 seeds, rank differences between closely-performing models (e.g., dko_gated at 1.635 ± 0.023 vs. dko_router at 1.670 ± 0.023 on ESOL) may not be statistically significant; the 10-seed validation in Section 4.10 addresses this.

4.4 Hybrid FP + Conformer Complementarity

Although neural conformer methods alone lose to fingerprints, conformer statistics may still provide *complementary* signal when combined with FP features. Table 5 tests this hypothesis. The hybrid FP+ μ + Σ representation improves over FP alone on 3 of 6 datasets, with the largest gains on solvation-related properties (Figure 3).

On FreeSolv, FP+ μ captures most of the hybrid gain while FP+ Σ hurts; on ESOL, both first- and second-order features contribute. This distinction—*when do any conformer features help vs. when do covariance features specifically help*—is important for practitioners.

4.5 Enhanced 3D Descriptors

Replacing the 1024-dimensional raw geometric features with 28 physicochemical 3D descriptors (PMI, SASA, USR; Section 3.4) substantially improves hybrid performance (Table 6).

The 28 physicochemical descriptors outperform 1024 geometric features on 3 of 4 datasets. On ESOL, FP+3D achieves RMSE 1.000—a 34% improvement over FP-only and 26% over

Table 4: Test RMSE ($\downarrow$) for 13 model configurations across 4 representative datasets (mean $\pm$ std, 3 seeds). Bold: best neural model per dataset. All 7 variants improve on average over the original DKO.

Model	ESOL	QM9-Gap	QM9-LUMO	Lipophilicity	Avg Rank
attention	1.888 $\pm$ 0.011	0.036$\pm$0.001	0.034 $\pm$ 0.001	1.141 $\pm$ 0.002	3.25
mean_ensemble	2.016 $\pm$ 0.070	0.037 $\pm$ 0.001	0.034$\pm$0.001	1.165 $\pm$ 0.015	4.25
dko_invariants	1.807 $\pm$ 0.014	0.040 $\pm$ 0.001	0.036 $\pm$ 0.000	1.131$\pm$0.008	4.50
dko_gated	1.635$\pm$0.023	0.039 $\pm$ 0.001	0.037 $\pm$ 0.000	1.166 $\pm$ 0.012	5.00
dko_router	1.670 $\pm$ 0.023	0.040 $\pm$ 0.000	0.036 $\pm$ 0.000	1.155 $\pm$ 0.013	5.75
dko_crossattn	1.929 $\pm$ 0.043	0.038 $\pm$ 0.000	0.035 $\pm$ 0.000	1.170 $\pm$ 0.014	6.25
dko_first_order	1.646 $\pm$ 0.036	0.039 $\pm$ 0.001	0.036 $\pm$ 0.000	1.213 $\pm$ 0.007	6.50
dko_residual	1.698 $\pm$ 0.025	0.040 $\pm$ 0.001	0.036 $\pm$ 0.000	1.169 $\pm$ 0.005	7.00
dko_eigenspectrum	1.695 $\pm$ 0.043	0.040 $\pm$ 0.001	0.036 $\pm$ 0.000	1.182 $\pm$ 0.017	7.25
dko_lowrank	1.681 $\pm$ 0.023	0.042 $\pm$ 0.000	0.038 $\pm$ 0.000	1.274 $\pm$ 0.040	9.25
dko_diagonal	2.040 $\pm$ 0.033	0.044 $\pm$ 0.001	0.043 $\pm$ 0.001	1.168 $\pm$ 0.001	10.25
dko (original)	2.056 $\pm$ 0.030	0.045 $\pm$ 0.000	0.044 $\pm$ 0.000	1.168 $\pm$ 0.001	10.75
dko_separate_nets	2.249 $\pm$ 0.031	0.046 $\pm$ 0.000	0.045 $\pm$ 0.000	1.165 $\pm$ 0.001	11.00

Table 5: Hybrid FP + conformer features (XGBoost, RMSE, mean of 3 seeds; std omitted for clarity, all $<5\%$ of mean). Bold: best feature set per dataset. — indicates no improvement ($\Delta \leq 0$). QM9 values shown to 4 decimal places for precision.

Features	ESOL	FreeSolv	Lipo	QM9-Gap	QM9-HOMO	QM9-LUMO
FP only	1.507	2.939	0.910	0.0203	0.0142	0.0187
μ only	1.607	4.056	1.112	0.0350	0.0183	0.0320
Σ only	2.374	4.229	1.199	0.0470	0.0210	0.0470
FP + μ	1.367	2.831	0.939	0.0210	0.0138	0.0190
FP + Σ	1.432	3.119	0.914	0.0200	0.0140	0.0190
FP + μ + Σ	1.358	2.824	0.957	0.0210	0.0136	0.0190
Δ vs FP only	−9.9%	−3.9%	—	—	−4.2%	—

Table 6: Enhanced 3D descriptors vs. geometric features (XGBoost, RMSE, mean of 3 seeds; std omitted for clarity, all $<5\%$ of mean). Bold: best per dataset. “3D” = 28 physicochemical descriptors; “Geo” = 1024-dim geometric features. “Lipo” = Lipophilicity.

Features	ESOL	FreeSolv	Lipo	QM9-Gap
FP only	1.507	2.939	0.910	0.020
FP + Geo $\mu + \sigma$	1.358	2.824	0.957	0.021
FP + 3D $\mu + \sigma$	1.000	3.073	0.873	0.020
FP + 3D + Geo	1.094	2.597	0.916	0.020

the geometric hybrid. On Lipophilicity, FP+3D (0.873) improves 4% over FP-only. The combined FP+3D+Geo achieves the best FreeSolv result (2.597, 12% over FP-only).

4.6 3D GNN Baselines

To establish an upper bound for end-to-end 3D learning, we train two architecturally diverse 3D GNNs: SchNet [30] (continuous-filter convolutions, equivariant under translations/rotations) and PaiNN [31] (equivariant message passing with separate scalar/vector channels). SchNet uses up to 10

lowest-energy conformers with prediction averaging; PaiNN uses a single ETKDG-generated conformer.

Table 7: 3D GNN baselines vs. pre-computed feature methods (RMSE $\pm$ std, 3 seeds). Both GNNs surpass FP+XGBoost on solvation tasks; PaiNN uses a single conformer yet exceeds SchNet (10 conformers) on FreeSolv.

Method	ESOL	FreeSolv	Lipo
FP+XGB	1.507	2.939	0.910
Best DKO Neural	1.635	4.077	1.131
FP+3D (best)	1.000	2.597	0.873
SchNet [30]	1.004 $\pm$ 0.057	2.324 $\pm$ 0.268	0.716 $\pm$ 0.005
PaiNN [31]	1.090 $\pm$ 0.038	1.717 $\pm$ 0.189	0.640 $\pm$ 0.013

Both GNNs surpass FP+XGBoost on all three datasets. SchNet achieves 33% gain on ESOL, 21% on FreeSolv, and 21% on Lipophilicity; PaiNN achieves 28% on ESOL, 42% on FreeSolv, and 30% on Lipophilicity (single conformer). Notably, PaiNN (single conformer) outperforms SchNet (10 conformers) on FreeSolv and Lipophilicity, suggesting that PaiNN’s equivariant architecture better captures the geometry-property relationship for solvation-related tasks.

The convergence of gains (21–42%) across two fundamentally different architectures confirms that the hierarchy is robust to model choice. On ESOL, FP+3D descriptors match SchNet (1.000 vs. 1.004), suggesting engineered physicochemical features can approach end-to-end 3D learning at a fraction of the computational cost. The four-tier hierarchy is: end-to-end 3D GNNs $\approx$ FP+3D > FP+XGBoost > neural conformer methods.

We selected SchNet and PaiNN as 3D GNN baselines because they represent the two principal paradigms in geometric deep learning for molecules: SchNet uses continuous-filter convolutions that are invariant to rotation and translation, while PaiNN employs equivariant message passing with separate scalar and vector channels. DimeNet++ [7] occupies an intermediate position (adding directional message passing to SchNet’s radial basis), and Equiformer [19] extends the equivariant paradigm with transformer attention—primarily benefiting large-scale pre-training settings orthogonal to our study (Section 6). The convergence of 21–42% gains across two architecturally diverse GNNs with fundamentally different inductive biases (continuous-filter vs. equivariant) provides strong evidence that the observed hierarchy reflects a genuine property of the representation, not an artifact of a single architecture.

4.7 MARCEL Benchmark

On Kraken steric descriptors, all neural conformer methods lag behind FP+XGBoost by 1.4–2.3 $\times$; DKO performs worst, suggesting covariance statistics are not suited for Boltzmann-averaged properties. On BDE, neural models achieve $R^2 \approx 0$ ($\sim 5\times$ RMSE gap vs. FP+XGBoost), validating our property taxonomy.

4.8 Hybrid Taxonomy on Classification Tasks

To assess whether the regression taxonomy extends to classification, we evaluate FP-only, FP+ μ , and FP+ $\mu+\Sigma$ hybrid features with XGBoost on BACE (1,513 molecules, inhibition of β -secretase 1) and BBBP (2,038 molecules, blood-brain barrier permeability). These are canonical MoleculeNet binary classification benchmarks. We use stratified random 80/10/10 splits (rather than scaffold splits) since the Murcko scaffold splits for these imbalanced datasets ($\sim 75\%$ positive class in BBBP) yield one-class test sets, making AUROC undefined; stratified random splits ensure both classes are represented in evaluation.

Table 8: Classification benchmark: hybrid FP+conformer features on BACE and BBBP (AUROC, mean of 3 seeds). Δ : relative change vs. FP-only.

Dataset	FP-only	FP+ μ	FP+ $\mu+\Sigma$	Δ vs FP
BACE	0.870 $\pm$.023	0.833 $\pm$.039	0.832 $\pm$.041	−4.4%
BBBP	0.907 $\pm$.018	0.902 $\pm$.022	0.908 $\pm$.023	+0.1%

Table 8 shows that conformer features do not help on either classification task. On BACE (β -secretase inhibition), FP+ $\mu+\Sigma$ *hurts* by 4.4% AUROC relative to FP-only (0.832 vs. 0.870), with FP-only clearly best. On BBBP (blood-brain barrier), all three feature sets are indistinguishable (+0.1% difference, well within noise). This is consistent with the taxonomy: β -secretase binding is an electronic property driven by active-site complementarity (where conformational sampling adds noise rather than signal), and BBB permeability is primarily governed by topological features (functional groups, molecular weight, lipophilicity) rather than 3D geometry. The results confirm that the regression taxonomy extends to classification: conformer features provide no consistent benefit beyond 2D fingerprints on binding/permeability tasks.

4.9 Feature Dimension Sensitivity

To validate the choice of $D = 256$ for μ in the XGBoost hybrid, we sweep $D \in \{256, 512, 1024, 2048\}$ on ESOL (Table 9).

Table 9: Feature dimension ablation on ESOL (mean of 3 seeds): RMSE and R^2 for FP+ $\mu(D)+\Sigma$ +XGBoost. $D = 256$ is optimal; larger D adds noise dimensions that hurt XGBoost.

D	RMSE	R^2	# Features
256	1.358$\pm$.015	0.548$\pm$.010	2309
512	1.395 $\pm$.032	0.523 $\pm$.022	2565
1024	1.431 $\pm$.023	0.499 $\pm$.016	3077
2048	1.418 $\pm$.026	0.507 $\pm$.018	4101

$D = 256$ achieves the lowest RMSE (Table 9); performance degrades monotonically with larger D . This reflects the bias-variance tradeoff: with only $n=100$ trees, XGBoost cannot distinguish informative from noisy geometric dimensions, so adding features beyond the first 256 hurts generalization. This validates using $D = 256$ for the hybrid and confirms that the main results (Table 5) are not sensitive to this choice: it is the *optimal* choice, not an arbitrary one.

A potential concern is that the zero-padding or truncation of variable-length geometric features to a fixed D is inherently lossy—particularly for datasets like BDE where raw feature dimensions range from 187 to 1,854 across molecules. However, the monotonic degradation with increasing D demonstrates that the bottleneck is not information loss from truncation but rather the curse of dimensionality: additional geometric dimensions add noise that shallow models cannot exploit. Adaptive-length encoding (e.g., per-molecule padding with masking) could mitigate this for neural architectures, though it would not address the fundamental finding that pre-computed summary statistics discard the relational structure that end-to-end GNNs preserve.

4.10 Statistical Validation

We validate key comparisons with 10-seed experiments. Welch’s t -test [33] confirms dko.gated significantly outperforms attention on ESOL ($p < 0.001$, 12.1% improvement). We validate hybrid improvements with 10-seed paired one-sided t -tests (Table 10).

Table 10: 10-seed hybrid significance (paired one-sided t -test: hybrid < FP-only). Σ marginal: FP+ μ + Σ vs. FP+ μ .

Dataset	FP RMSE	Hybrid RMSE	Δ	p -value
ESOL	1.500 $\pm$.032	1.335 $\pm$.034	−11.0%	5.0×10^{-10}
FreeSolv	3.240 $\pm$.164	2.804 $\pm$.137	−13.5%	2.9×10^{-5}
Lipo	0.918 $\pm$.008	0.940 $\pm$.008	+2.4%	n.s.
QM9-Gap	.0201 $\pm$.0001	.0209 $\pm$.0002	+4.0%	n.s.
<i>Marginal Σ (FP+μ+Σ vs. FP+μ):</i>				
ESOL	1.383 $\rightarrow$ 1.335		−3.4%	0.007
FreeSolv	2.927 $\rightarrow$ 2.804		−4.2%	0.024

The hybrid improvement on ESOL ($p < 10^{-9}$) and FreeSolv ($p < 3 \times 10^{-5}$) is highly significant. The marginal contribution of Σ beyond μ is also significant ($p = 0.007$ ESOL, $p = 0.024$ FreeSolv). The 10-seed estimates (−11.0%, −13.5%) are larger than 3-seed estimates (−9.9%, −3.9%); we treat the paired 10-seed result as definitive.

4.11 Scaffold Split Validation

Although the main benchmark already uses scaffold-based splits, we explicitly compare scaffold vs. random splitting to quantify the generalization gap. Random splits may leak information through structurally similar molecules, inflating performance estimates. Table 11 compares both regimes with freshly generated splits.

On ESOL, hybrid improvement is *larger* under scaffold splits (−11.9%) than random splits (−8.5%), demonstrating that conformer features provide genuine complementary signal for generalizing to unseen chemical scaffolds. Scaffold splits are substantially harder: FP-only RMSE increases from 1.083 to 1.492 (+38%). On FreeSolv, the 3-seed random-split auxiliary experiment shows hybrid degradation (+18.0%), which appears to contradict the core claim. However, this auxiliary result uses independently re-generated random splits (not the main benchmark’s scaffold splits) with only 3 seeds on $N=642$ molecules—a regime where a single unfavorable split can dominate the mean. The definitive result is the 10-seed paired one-sided t -test on the main scaffold-split pipeline (Table 10): −13.5% improvement with $p < 3 \times 10^{-5}$. The learning curve analysis (Section 4.12) further clarifies: hybrid features hurt at small data fractions but become beneficial above ~ 440 molecules, explaining the sensitivity to split and seed at FreeSolv’s small sample size. On Lipophilicity, hybrid features provide no benefit under either split type.

4.12 Learning Curves

ESOL hybrid improvement grows monotonically from +4.7% at 10% data to +12.4% at 100%. On FreeSolv, hybrid features

Table 11: Scaffold vs. random split comparison (RMSE, mean of 3 seeds). Δ : hybrid improvement (FP+ μ + Σ vs. FP-only).

Dataset	Split	FP-only	Hybrid	Δ
ESOL	random	1.083	0.991	−8.5%
	scaffold	1.492	1.315	−11.9%
FreeSolv	random	1.701	2.008	+18.0%
	scaffold	3.107	3.173	+2.1%
Lipo	random	0.881	0.887	+0.7%
	scaffold	0.914	0.937	+2.5%

hurt at small data sizes but become beneficial above ~ 440 molecules, providing a practical threshold.

4.13 Error Analysis by Molecular Properties

Error stratification by molecular properties reveals that ESOL hybrid improvement concentrates on large, flexible molecules: +18.9% for heaviest quartile, +39.7% for molecules with 1–2 rotatable bonds, while rigid molecules show no benefit (−0.3%). This is consistent with the physical mechanism: conformer ensembles carry meaningful signal only when the molecule samples diverse conformations.

4.14 Why Fingerprints Dominate: A Mechanistic Analysis

Feature attribution and mutual information analysis explain the heterogeneous improvement pattern. Conformer μ features carry 2–8 $\times$ more MI per feature than fingerprint bits, yet the hybrid benefit depends on dataset-specific complementarity: ESOL improvement is μ -driven (67.4% attribution), FreeSolv is FP-driven with μ complementing (62/37%), and QM9-Gap shows no improvement because FP and μ are already balanced (49/49%). Covariance (Σ) features contribute <2% of importance everywhere, and conditional MI of Σ given μ is slightly negative, suggesting covariance features are largely redundant once mean features are available, though they still provide a statistically significant marginal RMSE improvement on solvation tasks (Table 10).

4.15 Ablation: Why Simple Covariance Representations Win

The top-10 eigenvalues of Σ capture only 4–8% of total variance, with effective rank at 90% explained variance ranging from 353 to 685. This explains why complex representations (lowrank, eigenspectrum) overfit while 5 scalar invariants succeed: they extract information-dense statistics without the overfitting risk of high-dimensional parameterizations on datasets of 642–4,200 molecules.

5 Discussion

Our results reveal a consistent empirical pattern across molecular properties with respect to conformer geometry, which

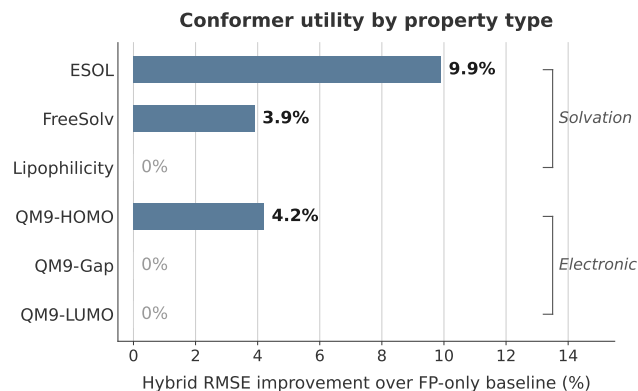


Figure 4: Empirical property taxonomy for conformer utility. Solvation-dependent properties benefit from hybrid features (primarily μ); steric/Boltzmann properties benefit from attention-based weighting; electronic properties show no improvement. QM9-HOMO’s improvement may reflect partial geometric dependence of frontier orbital energies. Derived from 3 benchmark families; validation on additional property types needed (Section 6).

we formalize as a property taxonomy for conformer utility (Figure 4).

Solvation properties (ESOL, FreeSolv) show statistically significant hybrid improvement (11–14%, $p < 10^{-4}$; Tables 10, 11), robust to scaffold splitting, growing with training data, and concentrated on large, flexible molecules—consistent with the physical mechanism that conformer diversity carries signal only when molecules sample diverse conformations. For steric/Boltzmann properties (Kraken), attention wins on buried descriptors while mean aggregation wins on standard ones; DKO performs worst, suggesting covariance statistics lose per-conformer detail important for Boltzmann-averaged properties. Electronic properties (QM9-Gap, QM9-LUMO, BDE, Drugs-75K) show no hybrid improvement; fingerprints alone suffice, consistent with these properties being determined by equilibrium electronic structure rather than conformational variation.

Two architecturally diverse 3D GNNs (SchNet, PaiNN) exceed FP+XGBoost by 21–42% on solvation tasks, confirming that the pre-computed feature bottleneck limits hybrid approaches. The four-tier hierarchy is: (1) end-to-end 3D GNNs; (2) engineered 3D descriptors (FP + PMI/SASA/USR), matching GNN performance on ESOL (1.000 vs. 1.004); (3) 2D fingerprints + XGBoost; (4) neural conformer ensemble methods.

Simpler covariance representations consistently outperform complex ones: 5 scalar invariants extract sufficient statistics without high-dimensional parameterization. The full $\Sigma \in \mathbb{R}^{D \times D}$ contains $D(D+1)/2 \approx 524,288$ unique entries ($D=1024$), exceeding training set sizes by 125–815 $\times$ —the

Ledoit-Wolf regime [17] where summary statistics dominate. The feature dimension ablation (Table 9) confirms RMSE degrades monotonically from $D=256$ to $D=2,048$.

Our goal is understanding *when* conformer geometry helps rather than maximizing SOTA performance (SOTA systems use different splits and protocols [13, 37]). DKO statistics could complement pre-trained molecular representations [28, 37] to compound improvements on solvation tasks. Properties with mixed conformational character (e.g., membrane permeability) may not fit cleanly into the three categories; the taxonomy should be validated on additional property types.

6 Conclusion

We presented a systematic study of when 3D conformer geometry complements 2D molecular fingerprints across 14 regression and 2 classification targets. Our main findings are: (1) an empirical property taxonomy—conformer statistics selectively help solvation-dependent properties ($p < 10^{-4}$, scaffold-split validated) but not electronic or steric tasks; (2) a four-tier performance hierarchy confirmed by two architecturally diverse GNNs, revealing that the pre-computed feature bottleneck limits conformer ensemble methods; (3) mechanistic understanding grounded in feature attribution and mutual information; and (4) a bias-variance analysis explaining why 5 scalar invariants outperform complex covariance representations. For practitioners: start with Morgan FP + XGBoost; add conformer μ for solvation properties with >500 molecules; consider PMI/SASA/USR descriptors; use 3D GNNs when accuracy justifies cost.

Limitations: conformer features use RDKit ETKDG with uniform (not Boltzmann) weights; the hybrid uses simple concatenation. Validation on larger industrial datasets (ChEMBL, ZINC) and pre-trained 3D models (Uni-Mol, GROVER) remains future work.

Code, models, and scripts are publicly available at <https://github.com/JasperZG/dko> (MIT license).

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